

Chapter 10

Gravimetric Methods of Analysis

Fundamentals of Analytical Chemistry, Skoog, et al., 10th Ed., 2022

Chapter 10 Problems:

10-10, 10-12, 10-18, 10-20

Read section 10A stop at A1, then read 10B.

Gravimetric methods of analysis are based on mass measurements with an analytical balance.

1. Volatilization Gravimetry: Analyte is converted to a gas and weighed
2. Precipitation Gravimetry

The analyte is converted to a sparingly soluble precipitate. The precipitate is then filtered, washed free of impurities, converted to a product of known composition by heat treatment, and weighed.

$$X\% = \frac{\text{mass } X}{\text{sample mass}} \times 100\%$$

$$X\% = \frac{\text{wt.ppt}(g) \times \frac{a}{b} \times \frac{\text{FW substance sought}(X) (g/mol)}{\text{FW ppt} (g/mol)}}{\text{sample mass} (g)} \times 100 \%$$

Where $\frac{a}{b}$ = gravimetric factor

a and b are never related to atoms of hydrogen or oxygen, instead look for atoms in the ppt and substance sought that are the same.

multiplying the factor in the numerator by the number of atoms of interest in the formula for the substance sought must equal the product of factor in the denominator times the number of atoms of interest in the ppt.

1. An iron ore was analyzed by dissolving a 1.1324 g sample in concentrated HCl. The resulting solution was diluted with water, and the iron (III) was precipitated as the hydrous oxide $\text{Fe}_2\text{O}_3 \cdot x \text{H}_2\text{O}$ by the addition of NH_3 . After filtration and washing, the residue was ignited at a high temperature to give 0.5394 g of pure Fe_2O_3 (159.69 g/mol). Calculate the % Fe (55.847 g/mol) in the sample.

2. A 20.2000 g sample of aluminum sulfate (FW = 342.14) was dissolved in water diluted to 250.0 mL and 50.0 mL aliquots taken for analysis. The sulfate was precipitated as barium sulfate (FW = 233.4) filtered off, dried, and weighed. If an aliquot treated in this manner produced 422.0 mg of BaSO_4 , calculate the % (w/w) aluminum sulfate in the original sample.

3. A 0.2795 gram sample of an insecticide containing DDT ($\text{C}_{14}\text{H}_9\text{Cl}_5$; FW = 354.5) was burned in a stream of oxygen in a quartz tube. The products (CO_2 , H_2O , and HCl) were passed through a solution of NaHCO_3 . After acidification, the chloride in this solution yielded 116.1 mg of AgCl (FW = 143.32). Calculate the percentage (w/w) DDT in the sample assuming that DDT is the only source of chloride.